EPFL

Practical Work: Assessment of Thin Si_3N_4 PVD Coatings

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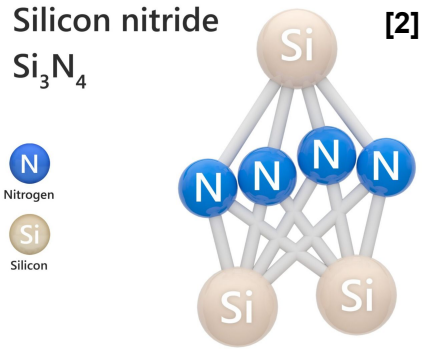
Introduction

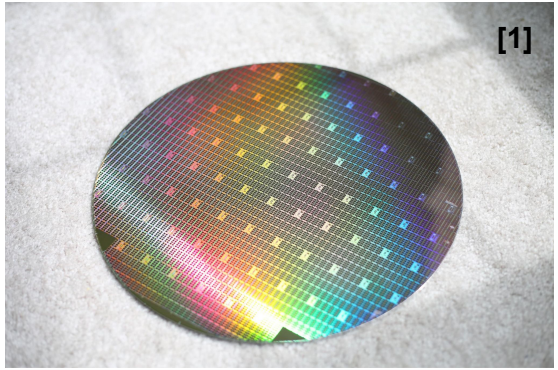
Silicon nitride (Si_3N_4) :

- Excellent mechanical, thermal, chemical, and electrical properties
- Widely used as an insulator layer in microelectronics



Context of the study:

- Two Si_3N_4 layers deposited on Si wafers
- Films show different electrical behavior and different coloration



Objectives of the Lab Work

Main goals:

1. Detect possible **carbon or oxygen contamination**

2. Determine the **Si/N stoichiometry** of the coating

3. Investigate the **Si_3N_4/Si interface** for contamination or deviations

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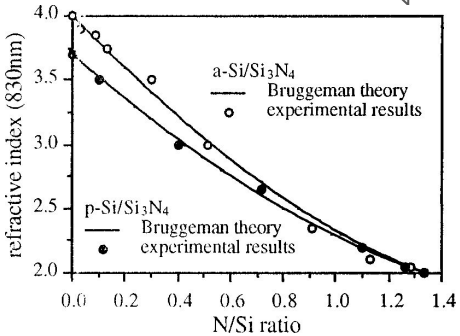

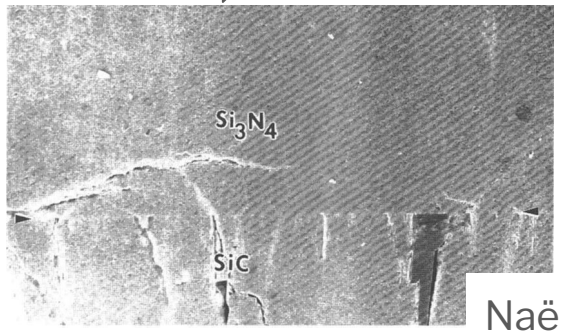

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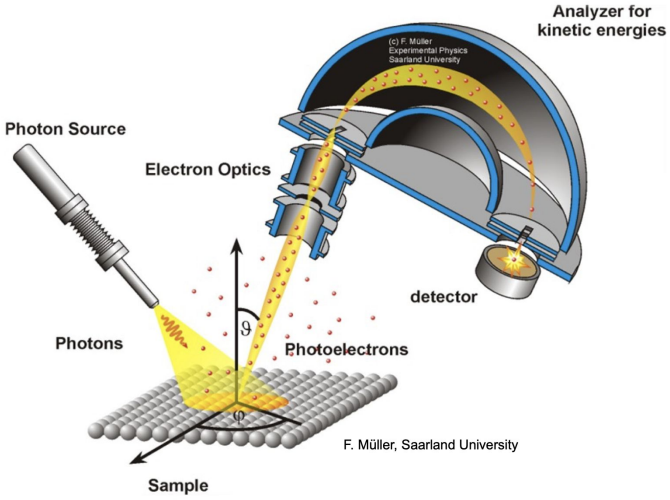
Characterization techniques

XPS: X-ray Photoelectron Spectroscopy

- Surface chemical composition
- Information on oxidation states
- Identification of contamination (C1s, O1s)
- Determination of Si/N ratio
- Low lateral resolution: 10-100 µm

AES: Auger Electron Spectroscopy

- Depth profiling
- [illegible]e contamination
- Chemical gradients across the coating
- Limited to conductive materials
- Much higher lateral resolution than XPS



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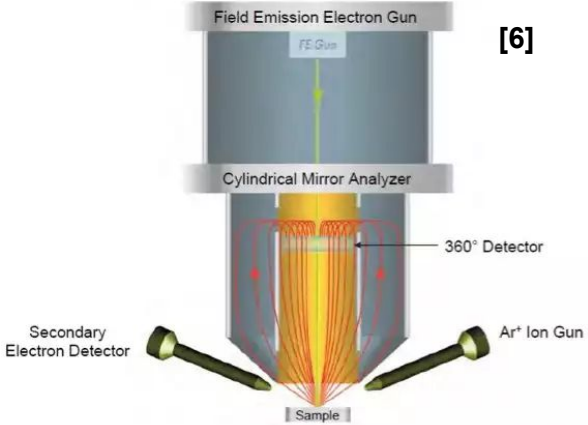


[6]

Schematic diagram of the cylindrical mirror analyzer with coaxial FE electron gun. The primary electron beam path is shown in green and the 360° pathway for detecting Auger electrons is shown in red.

Characterization techniques - Choice

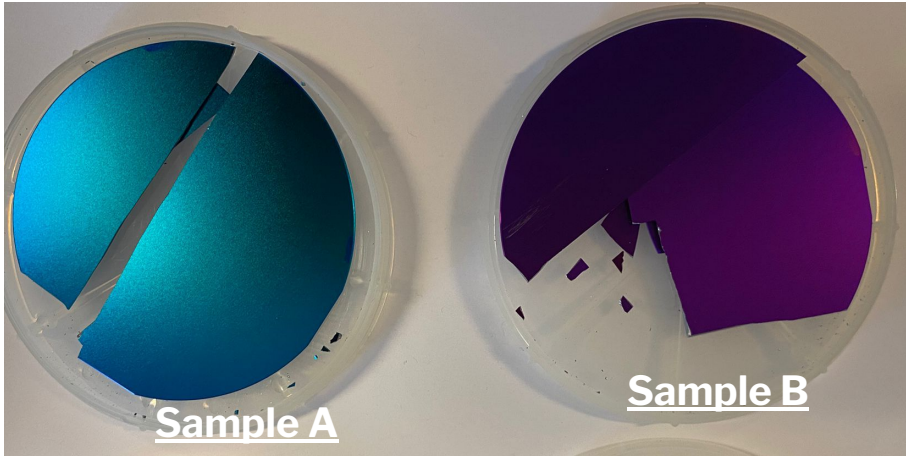
[7]

Mathelyo

Scientific Background - Hypothesis

- **Two initial samples:**
  - **Sample A:** green to blue color with high variation of intensity and color
  - **Sample B:** purple color with less variation and more dark intensity of color

=> two samples with different colors which may be explained.



- **Ideal composition:**
  - Stoichiometric silicon nitride: Si_3N_4
  - Expected N/Si atomic ratio ≈ **1.33**

- **Why contamination matters (and possible sources):**
  - Carbon → residue from process gases
  - Oxygen → oxidation or poor vacuum → can modify electrical insulation

- **Color variation can indicates differences in:**
  - Thickness
  - Composition
  - (Density or refractive index)

High resolution Spectra

Depth Profiling

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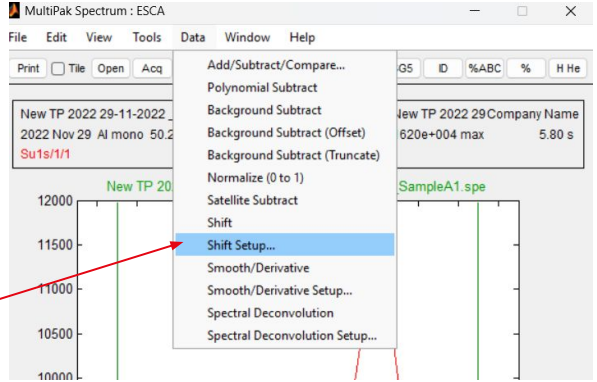
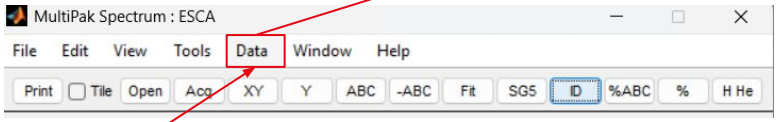
MultiPak

- **Using MultiPak Software to interpret the measured data**
  - ID the elements
  - Survey and High-Resolution Spectra
  - Atomic Concentration

- **<u>Survey Spectra Steps</u>**
  1. Import the .spe data file
  2. **Identify** the different elements
  3. Shift Carbon to peak (284.8 eV)





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XPS Results - Survey Spectra

- **Sample A:**

- Oxygen : 3.7 E4 c/s at 532.0 eV
- Nitrogen: 3.8 E4 c/s at 397.6 eV
- Carbon: 1.2 E4 c/s at 284.8 eV
- Silicon 2p: 2.3 E4 c/s at 101.5 eV

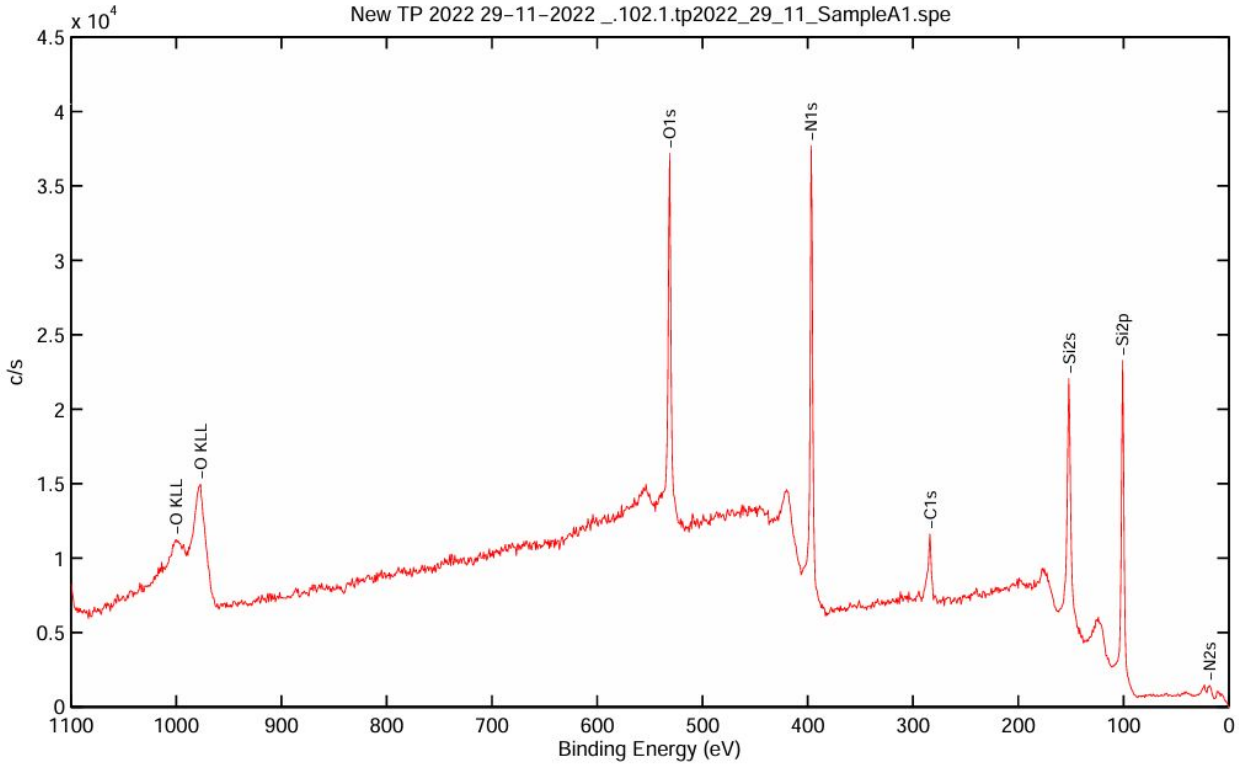

Figure 1 : XPS Survey Spectra of Sample A

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XPS Results - Survey Spectra

- **Sample B:**

- Oxygen : 3.8 E4 c/s at 532.4 eV
- Nitrogen: 4.7 E4 c/s at 397.2 eV
- Carbon: 1.0 E4 c/s at 284.5 eV
- Silicon 2p: 2.6 E4 c/s at 102.1 eV

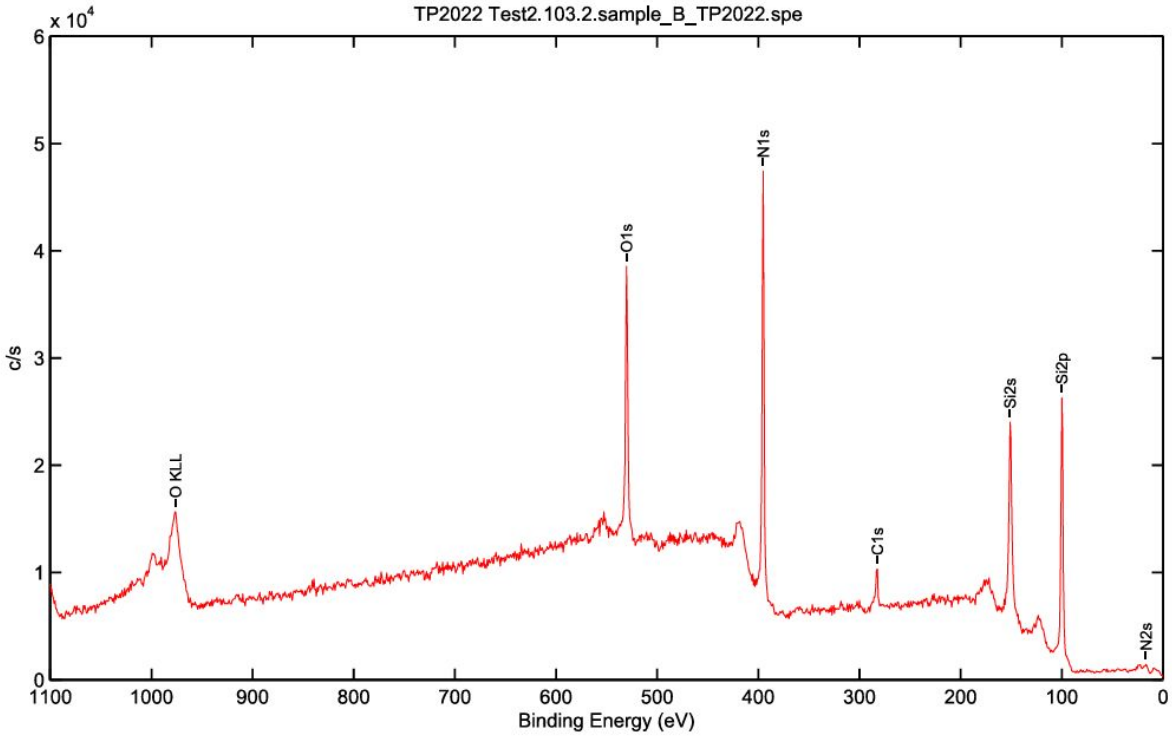

Figure 2 : XPS Survey Spectra of Sample B

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XPS Results - Survey Spectra

- **Comparison:**

Only difference seen on this graph is the content of nitrogen which is much higher for sample B.

Oxygen is detected in both samples → mainly attributed to surface oxidation

Carbon → contamination from air exposure

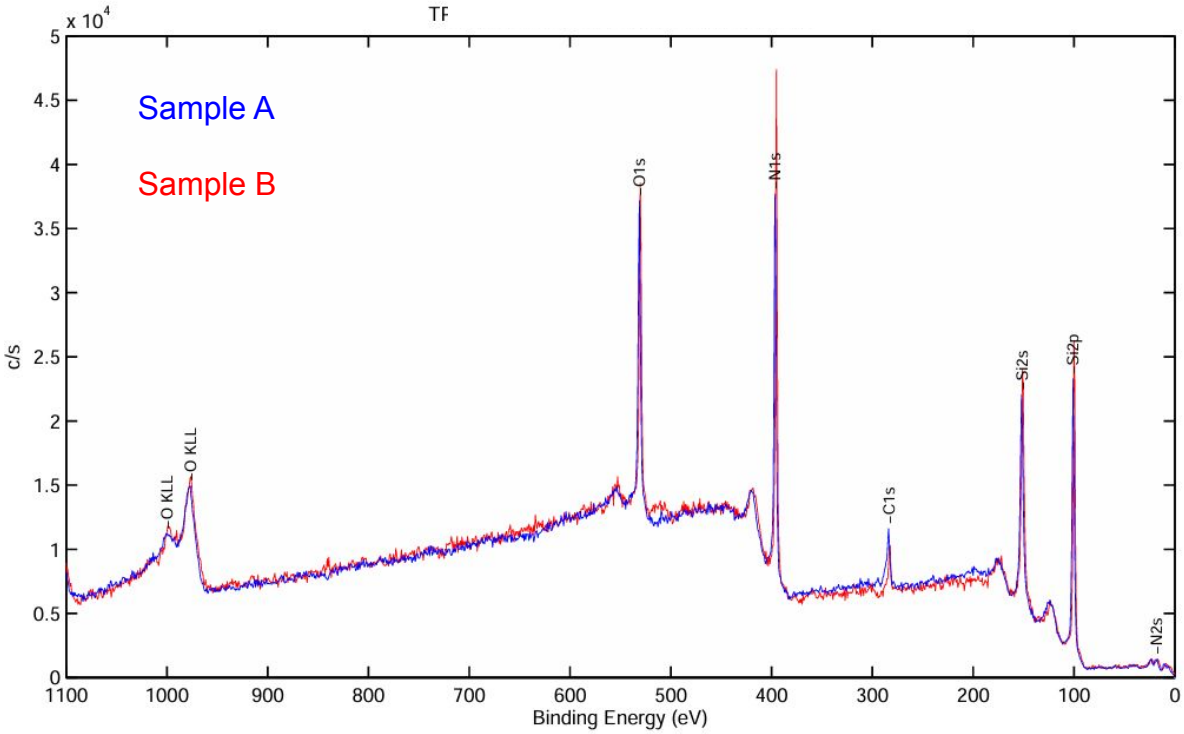

Figure 3 : Overlaid Survey Spectra (Samples A & B)

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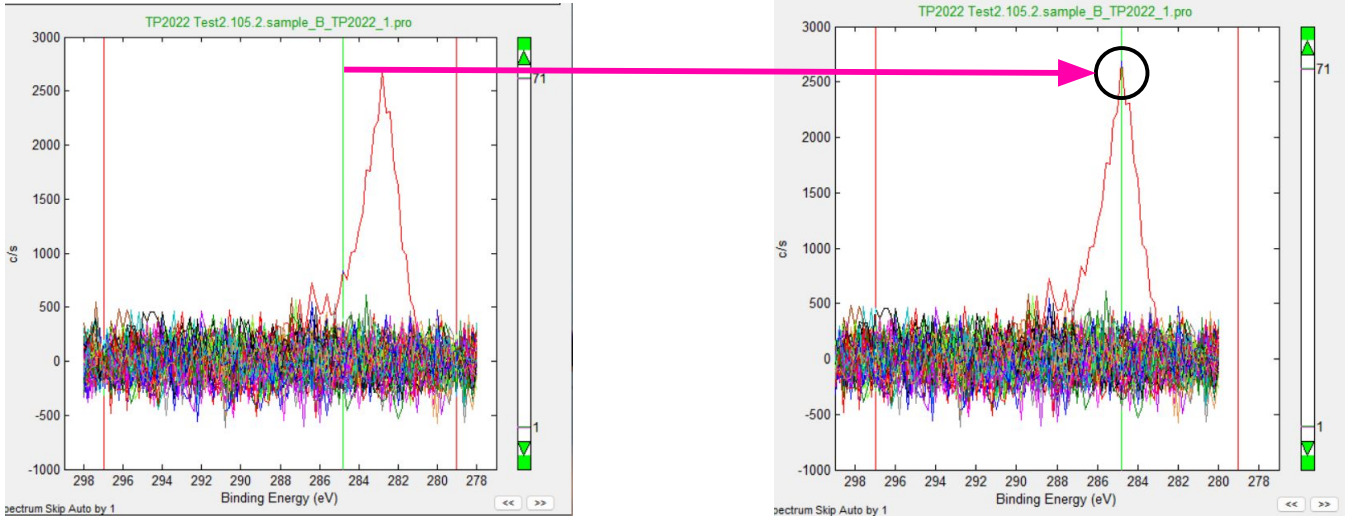
MultiPak - Depth Profiling

Depth profiling Steps

1. Export the .pro file
2. Background Subtraction for Carbon and Carbon shift



3. Background subtraction for other elements
4. Fix Window to reduce noise and avoid overlap with any neighbouring peaks

DON'T FORGET TO TURN ON DATALINK TO GET THE CORRECT ATOMIC CONCENTRATION

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XPS Results - Depth Profiling

Sample A:

- **Thickness:** N disappears
- **Surface contamination:** C and O
- **Interface contamination:** O

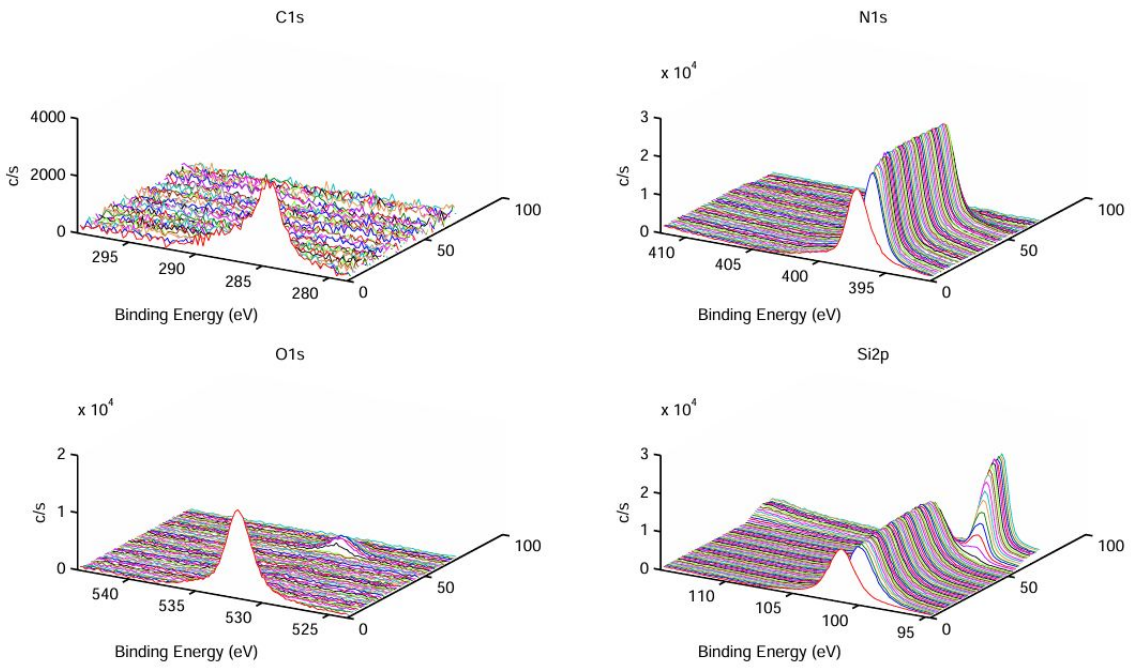

Figure 4 : XPS depth profiles for each element of PVD coating A on Silicon substrate

David

XPS Results - Depth Profiling

Sample B:

- Same impurity elements
- Thinner Film

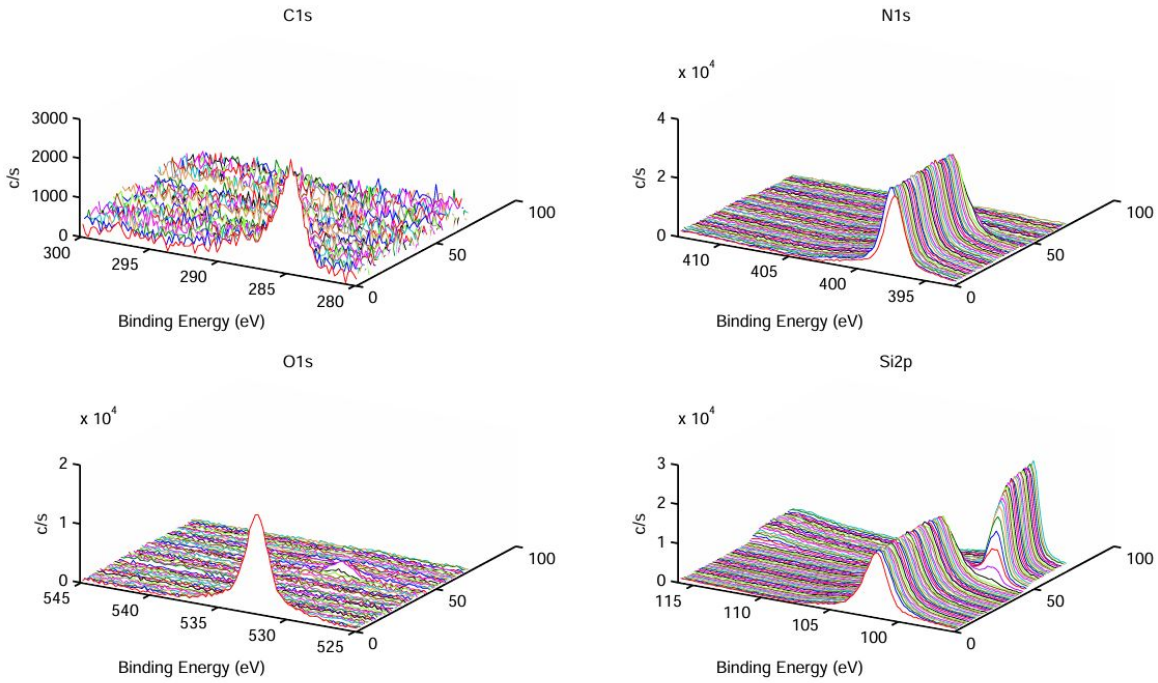

Figure 5 : XPS depth profiles for each element of PVD coating B on Silicon substrate

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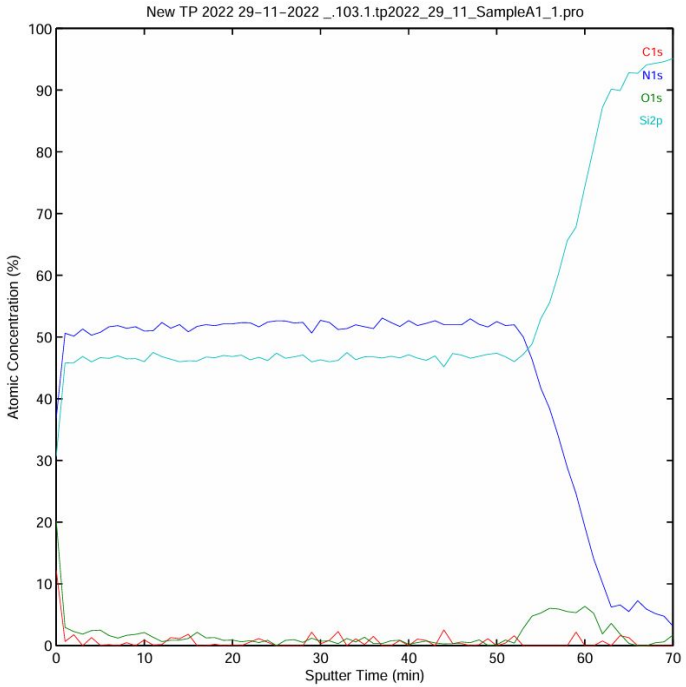
XPS Results - Interpretation

Atomic Concentration vs. Sputter Time

- Sample A:



- Sample B:

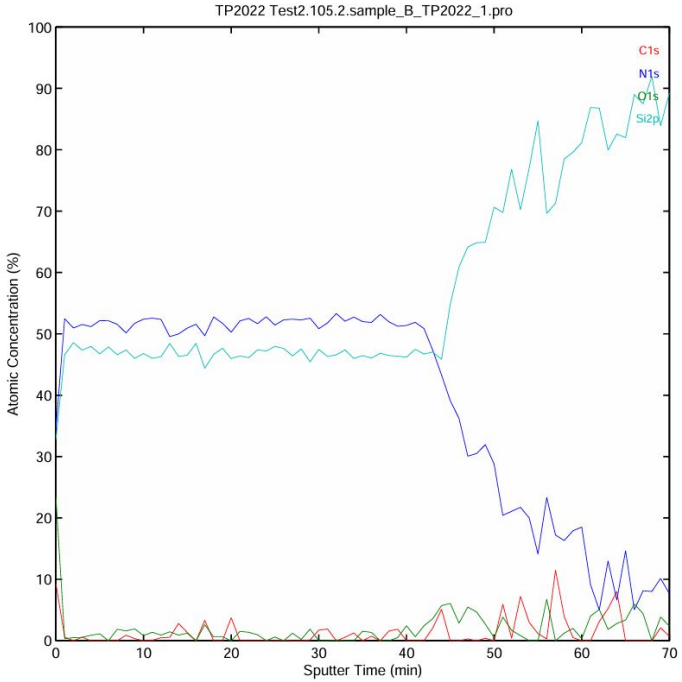


Figure 6 : Depth Profiles for Samples A & B

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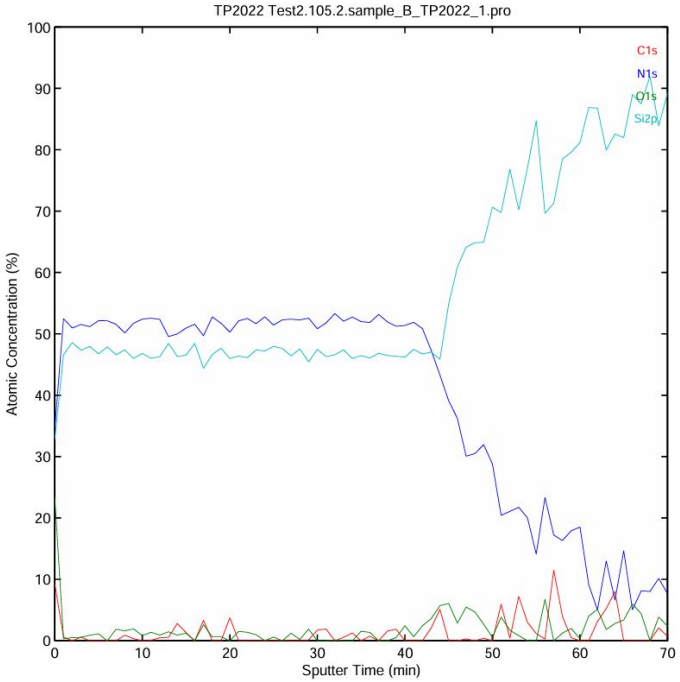
XPS Results - Interpretation

Atomic Concentration vs. Sputter Time

- Sample B:

Redeposition of interface contaminants



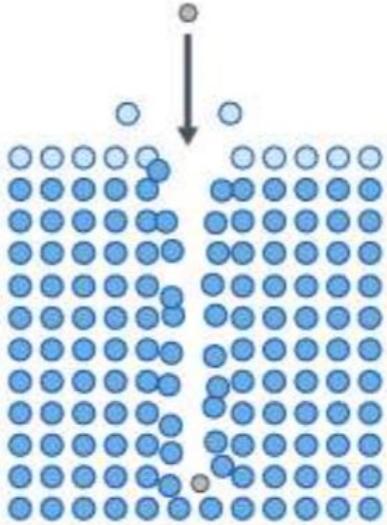


Figure 7 : Redeposition

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XPS Results - Interpretation

Atomic Concentration

	N [%]	Si [%]	Proportion N/Si
Ideal	57	43	1.33
Sample A	51.80	46.61	1.11
Sample B	52.35	46.42	1.13

Stoichiometry:

- Not Si_3N_4
- Ratio Sample A to Ideal = 0.834
- Ratio Sample B to Ideal = 0.849

→ **Stoichiometry is not the cause of the difference in color**

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XPS Results - Interpretation

Determination of the PVD coating thickness:

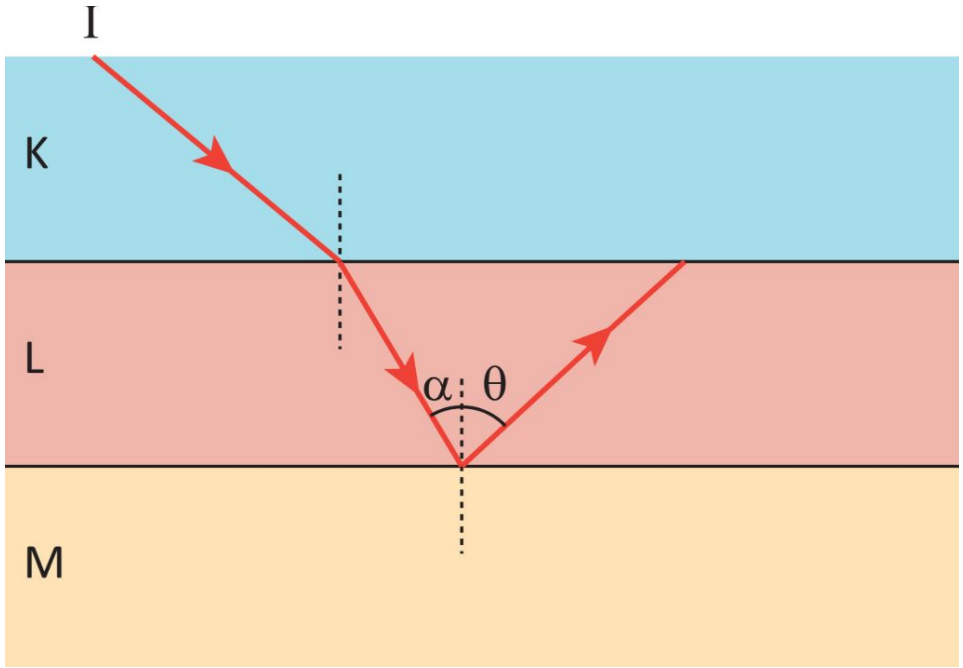
- Use the 50% drop rule (Nitrogen) [8]
- Sputter Rate ~ 8 to 9 nm / min, 1 cycle / min
- Examining the atomic concentration of Nitrogen in the depth profile.

	Max % / 2	Sputter Time [min]	Depth Range [nm]
Sample A	24.64	59	[472, 531]
Sample B	24.61	48	[384, 432]

XPS Results - Interpretation

Sample color:

- **Sample A** is thicker (~100nm thicker than sample B)
  - Absorbs more energy
  - Higher wavelength emitted



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Conclusion

- **Non-Stoichiometric Composition:** Both samples are Si-rich (N/Si ~ 1.1) and deviate from the ideal **Si_3N_4** ratio (1.33), yet they are chemically identical to each other. Stoichiometry is not the cause for the colour difference.
- PVD film thickness underlies the difference in colour (sample A is 100 nm thicker than sample B)
- **Sample A is the "Good" sample:** Exhibits a sharper coating-to-substrate transition, indicating a higher quality, distinct layer.
- **Sample B ("Bad"):** Shows a broader, less defined gradient caused by **redeposition of carbon contaminants** during sputtering.

THANK YOU

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